PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Prevalence and risk factors of frailty among people in rural areas: a systematic review and meta-analysis
AUTHORS	Xu, Rui; Li, qiufang; Guo, Feifei; Zhao, Maoni; Zhang, Luyao

VERSION 1 – REVIEW

REVIEWER	Beilby, Justin National Health and Medical Research Council of Australia Centre of Research Excellence Frailty Trans-disciplinary Research To Achieve Healthy Ageing, Torrens University
REVIEW RETURNED	20-Sep-2020

GENERAL COMMENTS	This is a useful systematic review but would do with some editing regarding use of english language. I have a number of comments. Introduction I would have liked some discussion about the reversibility of frailty (with some patients) and the need for the early identification of the pre frail. I would also suggest the table summarising the current reported prevalence in other international studies would help the reader. Results I am surprised that in the six risk factors identified poly pharmacy was not found to be another risk factor. Could the authors comment. Discussion The discussion is difficult to easily follow. I would suggest that the authors use specific headings that highlight the six risk factors. A key findings and next steps section maybe helpful. I would also be interested on the authors view (s) of where the best interventions would be to tackle this significant rural clinical issue. Finally the authors make a vague comment regarding different screening tools - having completed this review do they have view on what would the most appropriate tool.

REVIEWER	Lewis, Emma G Newcastle University, Population Health Sciences Institute
REVIEW RETURNED	02-Nov-2020

GENERAL COMMENTS	This is a valuable contribution to the field of research and focuses on a neglected population group. The quality of the analysis and conclusions drawn seem of a high standard, but there are frequent errors of English grammar which should be addressed before re-submission.

	- The reviewer provided a marked copy with additional comments. Please contact the publisher for full details.

REVIEWER	Carter, Ben King's College London, Biostatistics & Health Informatics
REVIEW RETURNED	27-Nov-2020

GENERAL COMMENTS	I congratulate the authors on this work- I have the following major concerns over the robustness of the findings: 1) Abstract- This does not include any detail of the methods and the conclusions do not reflect that the evidence is very variable and caution is needed in the interpretation! 2) Intro>Objectives- It is not the objective of a SR to carry out a meta analysis. This is one very small part in a SR and the RQ should be to assess the prevalence of rural frailty 3) Protocol. Was this registered? where is it available from, when was it drafted? 4) Methods - How was frailty measured, and what cut offs used for frailty and pre-frailty ? 5) Methods> missing data. How was this dealt with 6) Methods> How was rural defined 7) Methods > Risk of Bias. How as a study level RoB determined and where are the domains and study level RoB presented and defended 8) Methods > dealing with heterogeneity. How was this dealt with? why have the authors presented a pooled analysis with I-sqaured > 90%? This demonstrates that the prevalence estimate is very uncertain 9) Methods > subgroups to explore heterogeneity. This section is missing 10) Results. The frail versus pre-frail data has some errors in the table, eg Llano2019. Please revise 11) Describe the characteristics of the included studies in greater detail. In particular compare the definition of rural used throughout the included studies 12) Results> Frailty. Discuss how frailty was measured and the differences in tools and how they are categories and any restrictions on the usage of the tools (eg CFS is only validated in the > 65's) 13) Results > The pooled results need to to introduce subgroups that explain some of the heterogeneity as the current estimate is difficult to estimate due to the considerable heterogeneity 14) Results > after the pooled results are improved to explain some of the heterogeneity the findings should be interpreted with far more caution than at present

	15) Discussion, you should be comparing to other studies that have attempted to measure the prevalence of frailty - and include the following references The following is a pre-covid Frailty study at scale and SR https://academic.oup.com/ageing/article/48/3/388/5281115 https://academic.oup.com/ageing/article/47/6/793/5063797 The following is a UK based community study including frailty https://www.mdpi.com/2308-3417/5/4/88 16) State the weakness of the tools used that have been used to measure frailty in this setting- 17) State the impact of these findings on public health 18) Compare to the findings at the current time of the relevance of frailty and being frail in a peri-covid-19 era. Note the following reference https://www.thelancet.com/journals/lanpub/article/PIIS2468-2667(20)30146-8/fulltext

VERSION 1 – AUTHOR RESPONSE

Response to the comments of Reviewer#1
Comment No.1: Introduction: I would have liked some discussion about the reversibility of frailty (with some patients) and the need for the early identification of the pre-frail. I would also suggest the table summarizing the current reported prevalence in other international studies would help the reader.
Response: We had added the contents of reversibility of frailty, the need for the early identification of the pre-frail and the current reported prevalence in other international studies in discussion.
Comment No.2: Results: I am surprised that in the six risk factors identified poly pharmacy was not found to be another risk factor. Could the authors comment.
Response: We didn't extract the information about poly pharmacy from the included studies, so it is can't be analyzed as a risk factor.
Comment No.3: Discussion: The discussion is difficult to easily follow. I would suggest that the authors use specific headings that highlight the six risk factors. A key findings and next steps section may be helpful. I would also be interested on the authors view (s) of where the best interventions would be to tackle this significant rural clinical issue. Finally the authors make a vague comment regarding different screening tools – having completed this review do they have view on what would the most appropriate tool.
Response: We made a revision and discussed the six risk factors in sections. We suggested that a frailty screening mchanism, paying attention to the key groups, and nutrition, exercise, diet interventions may be useful for older people in rural areas. We added a comment about the screening tools used and we recommend using the Fried Phenotype (FP).

Response to the comments of Reviewer#2
Comment No.1: This is a valuable contribution to the field of research and focuses on a neglected population group. The quality of the analysis and conclusions drawn seem of a high standard, but there are frequent errors of English grammar which should be addressed before re-submission.
Response: We had checked our manuscript carefully and revised the errors.

Comment No.2: Please see attached documents for full comments.
Response: We had changed the term “elderly” to “older people” or “older adults” and revised the wrong words. As for the definition of a rural area, we can't give a unified definition, so we included studies as long as it clearly points out that it conducted in rural contexts.
Response to the comments of Reviewer#3
Comment No.1: Abstract- This does not include any detail of the methods and the conclusions do not reflect that the evidence is very variable and caution is needed in the interpretation!
Response: We had revised the abstract and pointed out the heterogeneity between studies in conclusions.
Comment No.2: Intro>Objectives- It is not the objective of a SR to carry out a meta- analysis. This is one very small part in a SR and the RQ should be to assess the prevalence of rural frailty
Response: We had revised the state of the objective. The objectives of this systematic review were to synthesize the pooled prevalence and risk factors of frailty in rural areas through conducting a meta-analysis.
Comment No.3: Protocol. Was this registered? where is it available from, when was it drafted?
Response: This meta-analysis was not officially registered.
Comment No.4: Methods - How was frailty measured, and what cut offs used for frailty and pre-frailty?
Response: We had added the content about the measurement of frailty. The measurement of frailty or pre-frailty was based on the assessment tools used. For example, the Fried phenotype contained 5 components (weight loss, exhaustion, weakness, slowness and low physical activity) and subjects who were positive for three or more components were defined as frailty, those with 1 or 2 component(s) were defined as pre-frailty. In the studies where frailty or pre-frailty were not defined using the FP, we considered the definition used by authors.

Comment No.5: Methods> missing data. How was this dealt with.
Response: We asked for the missing data by sending emails to the corresponding author of the included articles.
Comment No.6: Methods> How was rural defined.
Response: We considered studies conducted in people living in the rural areas. We found that there is no unified definition of rural areas by reviewing the literature. Some studies think that rural areas are the places with the relatively backward economic development, while others think that rural areas are far away from cities. Therefore, we will include studies as long as it clearly points out that it conducts in rural contexts.
Comment No.7: Methods > Risk of Bias. How as a study level RoB determined and where are the domains and study level RoB presented and defended.
Response: We used the criterion proposed by Loney et al which was designed to critically assess studies about the prevalence. Each study was assigned a summative quality score ranging from 0 to 8, which was obtained by scoring a point for each of: random sample or whole population, unbiased sampling frame (i.e. census data), adequate sample size (>300 subjects), measures were the standard, outcomes measured by unbiased assessors, adequate response rate (70%), confidence intervals, subgroup analysis, and study subjects described. Only studies that scored 3 or above were deemed eligible for inclusion and studies which met more than four criteria were thought to have adequate quality.
Comment No.8: Methods > dealing with heterogeneity. How was this dealt with? why have the authors presented a pooled analysis with I-sqaured > 90%? This demonstrates that the prevalence estimate is very uncertain.
Response: The I-sqaured > 90% was calculated by the software of Stata 12.0 and we conducted stratified meta-analyses in order to explore the potential reasons for heterogeneity.
Comment No.9: Methods > subgroups to explore heterogeneity. This section is missing.
Response: We conducted the stratified meta-analyses and tried to explore the potential reasons for heterogeneity. In stratified meta-analyses, the literature data were divided into subgroups

according to frailty criteria, gender, and level of development, and pooled estimates of frailty prevalence with 95% CIs were calculated.
Comment No.10: Results. The frail versus pre-frail data has some errors in the table, eg Llano2019. Please revise.
Response: We had checked the data of Llano2019 and didn't find the errors of frailty and pre-frailty.

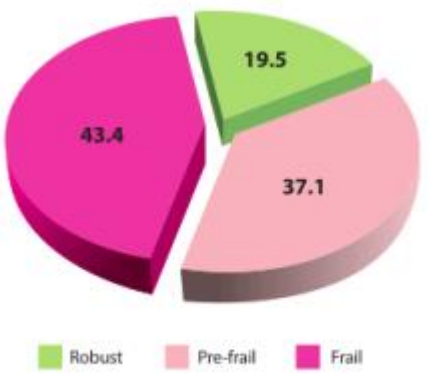


Figure 2 – Prevalence of self-reported frailty in the rural elderly, Pelotas, Rio Grande do Sul, Brazil, 2014 (N = 820)

Comment No.11: Describe the characteristics of the included studies in greater detail. In particular compare the definition of rural used throughout the included studies.
Response: We added the information about the study design, mean or median age of the population and proportion of female of the included studies. We didn't compare the definition of rural used throughout the included studies because most of the included studies didn't make the definition of rural.
Comment No.12: Results> Frailty. Discuss how frailty was measured and the differences in tools and how they are categories and any restrictions on the usage of the tools (eg CFS is only validated in the > 65's)
Response: We had added the content of comment about the assessment tools. The benefits of the FP include its relative ease of application, and another benefit is the minimal amount of data required for its calculation, making it a potential tool for screening. However, the FP has its roots in Canadian population surveys, and it is better to verify its applicability or to make an adjustment when used in other populations. The FI is another commonly used criterion, based on a deficit accumulation model, and study found that the eFI (the electronic Frailty Index, eFI) overestimates the frailty status of community dwelling older people compared with the Clinical Frailty Scale which is a validated measure of frailty based on clinical presentation. Hence, we need to consider carefully when using the FI and further work should also be conducted to identify a uniform scale that can be used to accuratelyidentify frailty among populations.
Comment No.13: Results >The pooled results need to introduce subgroups that explain some of the heterogeneity as the current estimate is difficult to estimate due to the considerable heterogeneity
Response: We had added the content of stratified meta-analyses and made a discussion according to the results of stratified meta-analyses.
Comment No.14：Results > after the pooled results are improved to explain some of the heterogeneity the findings should be interpreted with far more caution than at present.
Response: We had made a state that we should be cautious about the research results due to the heterogeneity between studies.
Comment No.15: Discussion, you should be comparing to other studies that have attempted to measure the prevalence of frailty - and include the following references
The following is a pre-covid Frailty study at scale and SR
https://academic.oup.com/ageing/article/48/3/388/5281115
https://academic.oup.com/ageing/article/47/6/793/5063797
The following is a UK based community study including frailty
https://www.mdpi.com/2308-3417/5/4/88

Response: We had made an attempt to compare to other studies and included below studies.
Comment No.16: State the weakness of the tools used that have been used to measure frailty in this setting.
Response: We had added the state of the weakness of the tools used. The benefits of the FP include its relative ease of application, and another benefit is the minimal amount of data required for its calculation, making it a potential tool for screening. However, the FP has its roots in Canadian population surveys, and it is better to verify its applicability or to make an adjustment when used in other populations. The FI is another commonly used criterion, based on a deficit accumulation model, and study found that the eFI (the electronic Frailty Index, eFI) overestimates the frailty status of community dwelling older people compared with the Clinical Frailty Scale which is a validated measure of frailty based on clinical presntation. Hence, we need to consider carefully when using the FI and further work should also be conducted to identify a uniform scale that can be used to accurately identify frailty among populations.
Comment No.17: State the impact of these findings on public health.
Response: The urban-rural gap is the main problem in the distribution of medical resources in in various countries and there are a large number of older people in rural areas who are the key populations to achieve equalization of basic public health services. As a relatively neglected elderly group, a better insight of frailty and its risk factors will have wide-scale implications for health care policy and practice in rural dwelling populations. This meta-analysis found that the pooled prevalence of frailty of older people was 18% in rural areas and many factors were associated with frailty, including age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health. Hence, it's time to focus the frailty of older people in rural areas. In order to prevent and reduce the frailty of the older adults in rural areas, we can try to establish a frailty screening mechanism for the elderly in rural areas, pay attention to the key groups, and establish nutrition, exercise, diet and other interventions.
Comment No.18: Compare to the findings at the current time of the relevance of frailty and being frail in a peri-covid-19 era. Note the following reference
https://www.thelancet.com/journals/lanpub/article/PIIS2468-2667(20)30146-8/fulltext
Response: We had discussed the frailty in the context of covid-19 and included this study.
We tried our best to improve the manuscript and made some changes in the manuscript. These changes will not influence the content and framework of the paper. And here did not list the changes but marked in red in revise paper.
We appreciate for Editors and Reviewers' warm work earnestly, and hope that the correction will meet with approval.
Once again, thank you very much for your comments and suggestion.

VERSION 2 – REVIEW

REVIEWER	Beilby, Justin National Health and Medical Research Council of Australia Centre of Research Excellence Frailty Trans-disciplinary Research To Achieve Healthy Ageing, Torrens University
REVIEW RETURNED	13-Jan-2021

GENERAL COMMENTS	This is a useful paper and should be published. I have a couple of comments. Abstract The authors make a blunt statement - " There was no publication bias". How can they be so definitive ? Should this be removed from the Abstract and added more subtly in the discussion. Introduction no comment

	Methods no comment Risk Factors I note the comments re risk factors but there is no mention of health system issues for rural people. Could this not identified /measured? Discussion The discussion is logical but dense and difficult to read. I would have preferred the use of section headings e.g. Prevalence ; Risk Factors; Health service issues, Conclusion

REVIEWER	Carter, Ben King's College London, Biostatistics & Health Informatics
REVIEW RETURNED	04-Jan-2021

GENERAL COMMENTS	The authors have improved the manuscript